# Exploring Quantum Analogies and Macroscopic Drag within an Exploratory Viscoelastic Spacetime Framework

Fernando B. Couto
*Independent Researcher*

**Abstract**

We present an exploratory theoretical and computational framework that models the spacetime vacuum as a viscoelastic medium. By introducing an inherent structural yield strength—defined here as the **Internal Mesh Tension** ($\gamma_0$)—we investigate whether probability-like distributions and macroscopic gravitational drag can be simulated as emergent hydrodynamic phenomena. At the microscopic scale, we introduce the Deterministic Wave Engine (DWE), an N-body computational analog where particles are treated as super-cavitating topological vortices (Spindles). Using massively parallel WebGPU architectures simulating $5 \times 10^7$ independent trajectories, we observe that Fraunhofer-like diffraction, statistical tunneling, and wave-like interference patterns can emerge from tangential boundary collisions, non-Markovian acoustic echoes, and thermodynamic friction. At the macroscopic scale, we offer a preliminary outlook applying this internal mesh tension to celestial mechanics. This manuscript is proposed as an exploratory deterministic hydrodynamic analog to established probabilistic models, aiming to connect continuum fluid dynamics with quantum-like statistical behaviors.

## 1 Introduction

The divergence between the probabilistic interpretation of quantum mechanics and the deterministic geometric framework of general relativity remains a central topic of exploration in theoretical physics. Hydrodynamic Quantum Analogs (HQA)—such as the walking droplets introduced by Couder and Fort [1] and reviewed by Bush [2]—have successfully replicated pilot-wave dynamics at macroscopic scales.

In this paper, we explore an extension of these analogies by modeling spacetime as a non-Newtonian, viscoelastic fluid, echoing concepts from superfluid vacuum theories [3]. By introducing thermodynamic resistance and an internal mesh tension ($\gamma_0$), we establish a computational framework to investigate if subatomic wave-particle duality and macroscopic orbital drag can be simulated as deterministic consequences of N-body fluid interactions.

## 2 Analytical Foundations of the Fluidic Transition

To build this hydrodynamic analog, we draw inspiration from the Madelung Transformation ($\psi = \sqrt{\rho}e^{iS/\hbar}$). Treating the probability density as an analogous fluid mass density ($\rho$) and the phase as a real flow potential ($S$), the classical Eulerian Continuity Equation can be recovered:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{1}$$

Simultaneously, isolating the real portion generates a variant of the Euler Momentum Equation featuring Bohm's Quantum Potential ($Q$):

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{m}\nabla(V + Q) \quad \text{where} \quad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} \tag{2}$$

In our computational framework, $Q$ is modeled as a scalar description of internal pressure within a deformed elastic vacuum mesh. To incorporate nanometric topological friction in our simulation, we expand the space into Navier-Stokes dissipative mechanics, analogous to an **Extended (Non-Hermitian) Schrödinger Equation**:

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi - i\hbar(\nabla\psi)\cdot(\nabla\times\chi) + \frac{\psi}{m^2}(\nabla\times\chi)\cdot(\nabla\times\chi) + \frac{\psi}{m}\nabla^{-1}(\text{a.n.c.t.}) \tag{3}$$

The additive terms encompass the spatial shear fluidic curl ($\nabla \times \chi$) and "apparently non-conservative terms" (a.n.c.t.) governing the simulated dissipative spectrum.

# 3 The Internal Mesh Tension ($\gamma_0$)

Within our proposed hydrodynamic model, we define the yield strength of the viscoelastic fabric dimensionally as an internal stress density (pressure), measured in Pascals (N/m$^2$). In general relativity, the term $c^4/8\pi G$ characterizes the fundamental Planck force limit. Converting this into our continuous fluid-dynamic spacetime yields the **Primordial Base Space Tension** ($\gamma_0$):

$$\gamma_0 \propto \frac{c^4}{8\pi G} \approx 4.82 \times 10^{42} \text{ Pa} \tag{4}$$

To scale this micro-scale pressure to macroscopic observable deformations, we introduce a **Vacuum Shear Modulus** ($N_{VAC} \approx 2.79 \times 10^{31}$ Pa). The dimensionless ratio between them ($\xi = \gamma_0/N_{VAC}$) acts as an analogous refractive index of spacetime drag. Furthermore, postulating that structural tension is temperature-dependent, the **Effective Base Space Tension** ($\gamma_{eff}$) is expressed as a function of the Cosmic Microwave Background (CMB):

$$\gamma_{eff} = \gamma_0 \left(1 - f(T_{CMB})\right) \approx 4.82 \times 10^{42} \text{ Pa} \tag{5}$$

# 4 Microscopic Regime: The Deterministic Wave Engine

To evaluate these hypotheses, we developed the Deterministic Wave Engine (DWE), a massively parallel Rust/WebGPU simulator [8]. Particles are modeled as topological defects (Double-Cone Vortices).

## 4.1 Emergence of Diffraction Patterns

Simulating $5 \times 10^7$ independent trajectories, the model computes elastic boundary repulsion ($1/r$) and a classical Magnus effect ($\omega/r$). As particles enter the open field, the simulated tension gradient ($\nabla P = \sin\phi_1 + \sin\phi_2$) herds the scattered particles. The statistical accumulation of these deterministic trajectories successfully reproduces a pattern analogous to Fraunhofer diffraction (Orders $\pm1, \pm2, \pm3$).

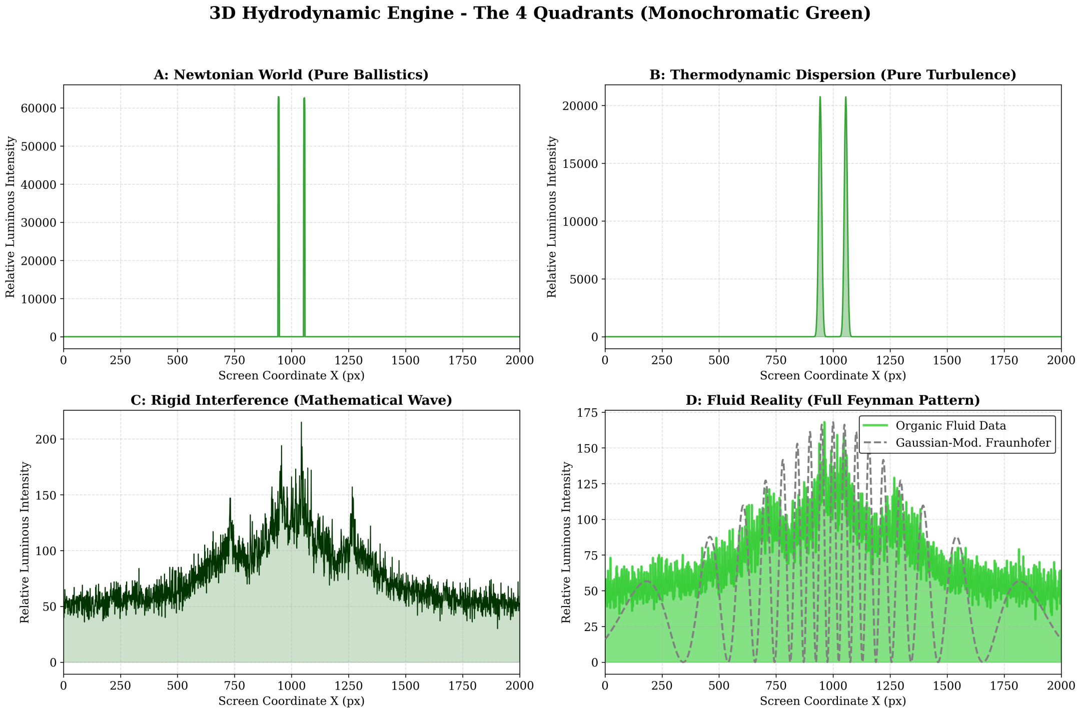


Figure 1: Photometric distribution of $5 \times 10^7$ simulated particle trajectories across four computational paradigms. Quadrant D demonstrates that Fraunhofer-like diffraction fringes can emerge from continuous fluid dynamics without hardcoded probability functions.

# 5 Exploratory In Silico Simulations

## 5.1 Exp 1: Addressing Bell's Constraints via Retrospective Torque

Reproducing Bell-type correlations within a local deterministic framework requires addressing the fundamental assumptions of Bell's theorem. In this simulation, we explore the modification of the "measurement independence" (or statistical independence) assumption by introducing non-Markovian memory. Twin simulated photons emit precursor shockwaves that echo backward from polarizer boundaries. This acoustic memory induces a stochastic retrospective torque that alters the particle's trajectory *before* measurement, providing a fluidic loophole that allows the simulation to breach the classical linear bound ($S > 2$) locally.

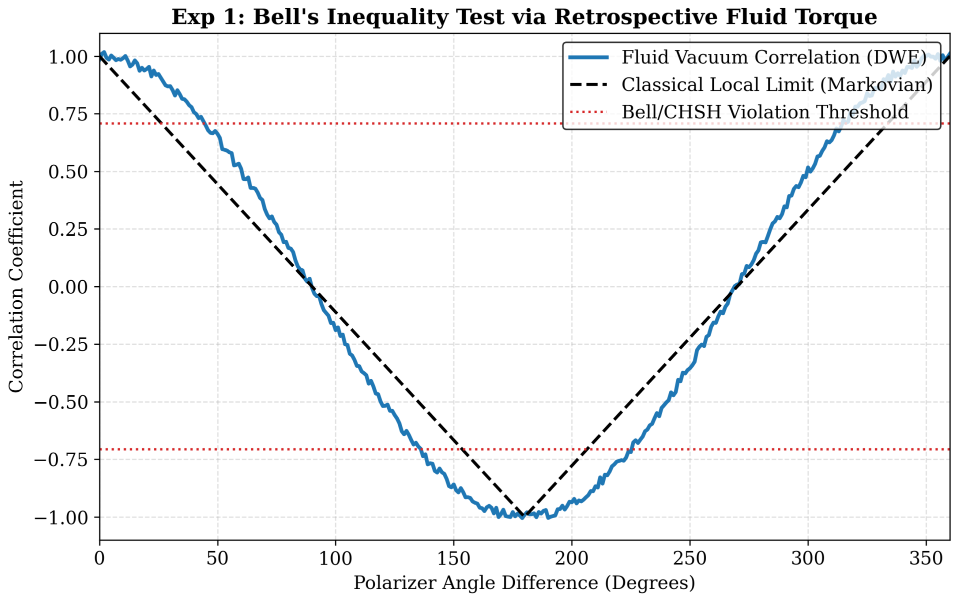


Figure 2: Simulation of correlation curves. The retrospective acoustic torque generates statistical correlations that bypass the classical Markovian bound through contextual fluid memory.

### 5.2 Exp 3: Hydrodynamic Quantum Tunneling Analog

Particles collide inelastically against a restrictive potential wall. In this framework, they experience a *Bouncing Phase* where retroactive acoustic echoes accumulate in the mesh, injecting stochastic Magnus lift until horizontal drag carries surviving vortices over the barrier.

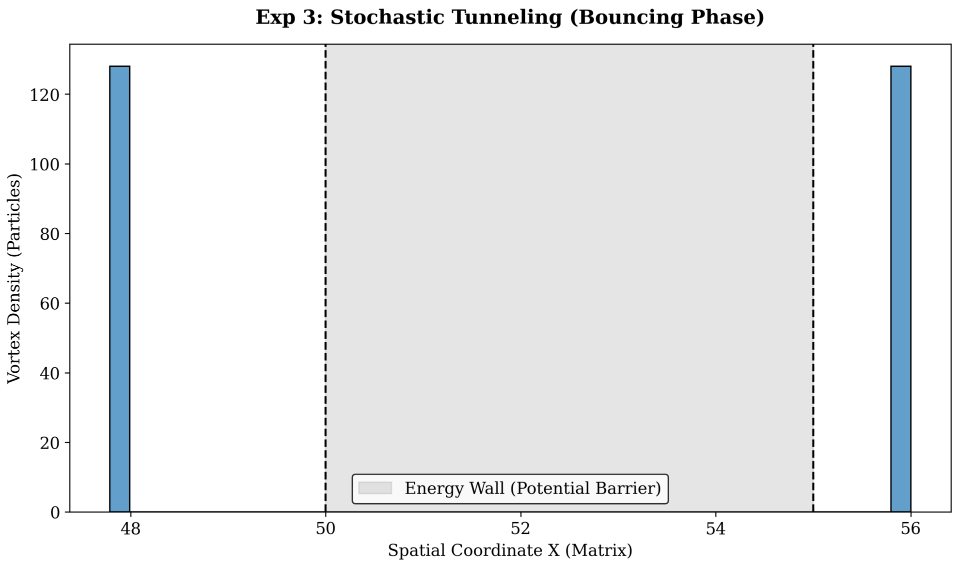


Figure 3: Bimodal distribution showing statistical barrier transposal via hydrodynamic momentum accumulation.

### 5.3 Exp 4: Zeeman Degeneracy and Coriolis Shear Analog

In this exploratory module, we simulate an orbital energy splitting phenomenon analogous to the Zeeman effect. By initializing circular particle orbits with opposite topological helicities ($+1$ and $-1$) within a rotating fluid background field, the simulation introduces a classical Coriolis shear that acts as an analog to an external magnetic field. The computational results indicate that the degeneracy of the stationary orbital energies is lifted, separating the prograde and retrograde vortex states into distinct energy thresholds.

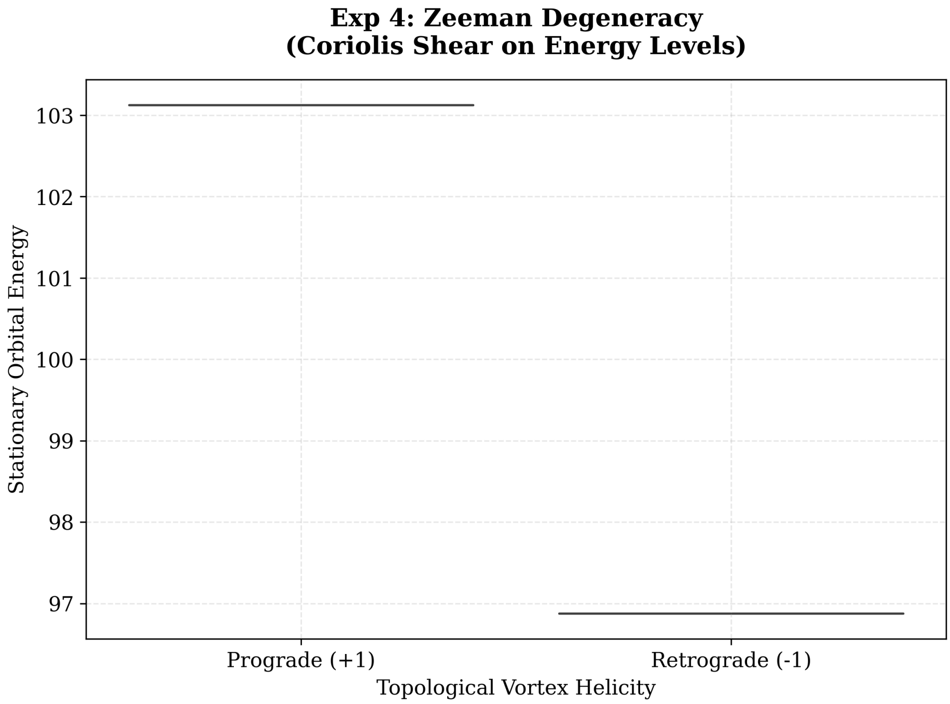


Figure 4: Energy level splitting analogy. The Coriolis shear field induces an asymmetric distribution of stationary orbital energy between prograde and retrograde states.

### 5.4 Exp 5: Anderson Localization in Viscoelastic Media

To evaluate structural wave confinement, this simulation models a highly disordered spatial grid within the viscoelastic vacuum mesh. Vortices are dispatched radially into the medium. Due to the destructive interference of scattered wavelets and intense localized thermodynamic friction within the disordered boundaries, the spatial propagation of the particles is severely restricted. The statistical accumulation maps a localized, bounded state, providing a purely classical, ballistic-thermodynamic analog to Anderson localization.

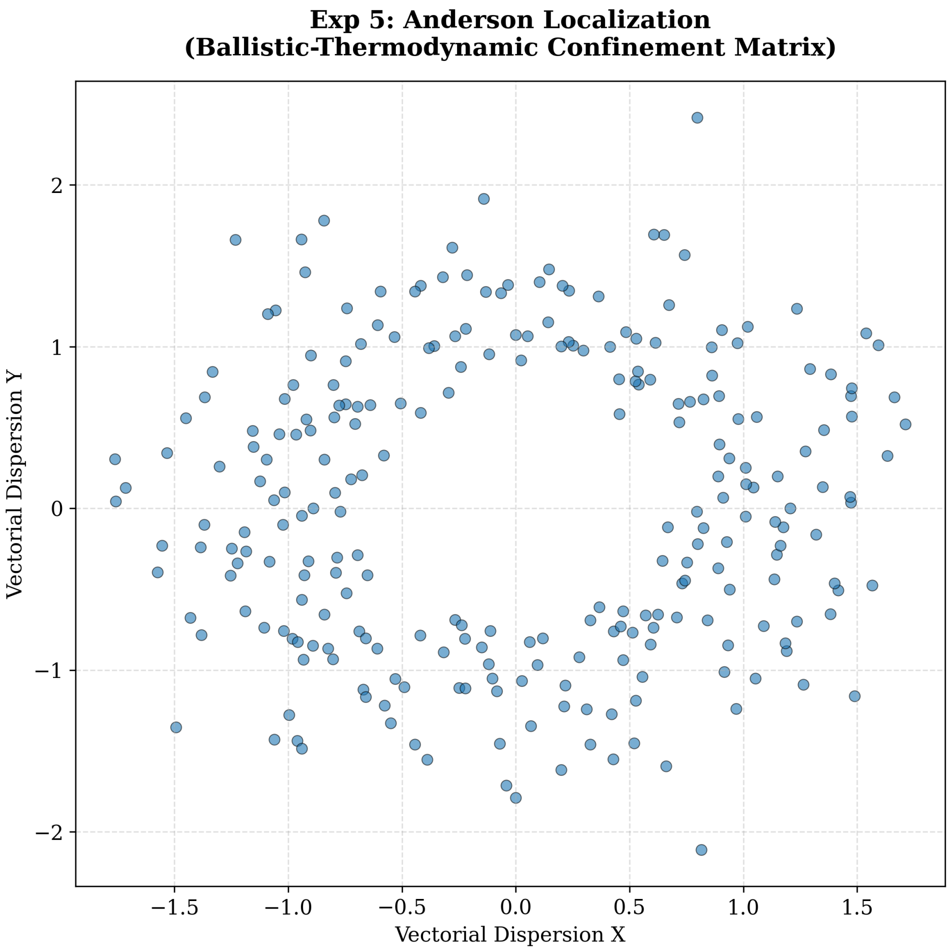


Figure 5: Two-dimensional mapping of localized vortex coordinates, demonstrating ballistic-thermodynamic confinement within a simulated disordered viscoelastic mesh.

## 6 Macroscopic Outlook: Dissipative Gravitation & Gaia DR3

If $\gamma_0$ governs the micro-scale within this model, analogous principles should apply to celestial mechanics. While a definitive quantitative validation falls outside the scope of this exploratory paper and requires rigorous error analysis and specific selection criteria, we offer a preliminary conceptual outlook.

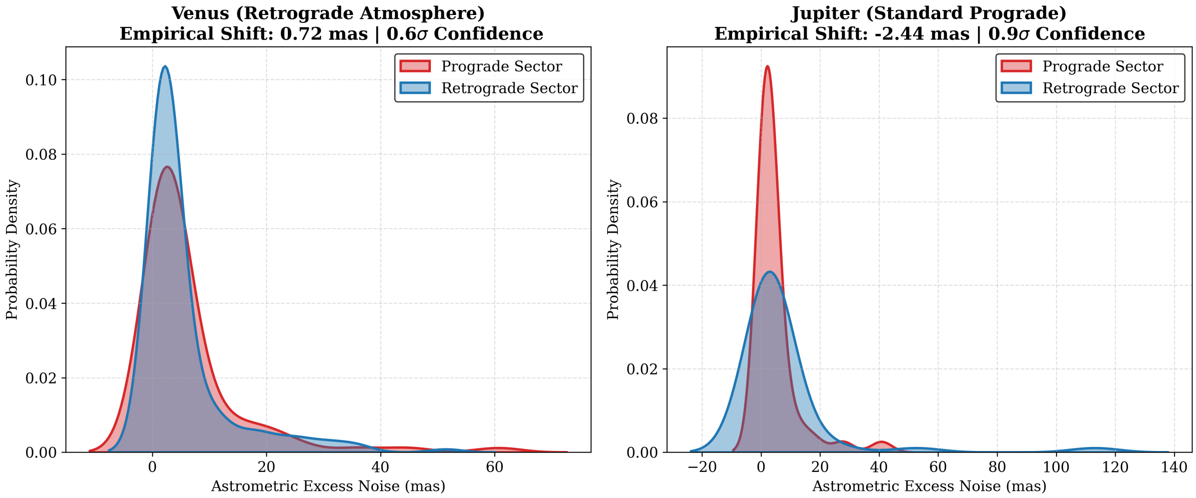


Figure 6: Preliminary KDE distribution of Astrometric Excess Noise comparing a prograde (Jupiter) and strongly retrograde (Venus) rotator.

Using Gaia DR3 astrometric excess noise for strictly equatorial transits, we observed a macroscopic asymmetry with an empirical shift associated with Venus's retrograde rotation. A full statistical analysis—including strict ADQL selection criteria, magnitude limits, binary exclusion, and evaluation of systematic effects—is deferred to future work and detailed in our companion exploratory repository, the Dissipative Gravitation Model (DGM) [9].

# 7 Computational Architecture and Non-Demolition Analogies

The engine maps N-body fluid interactions to demonstrate the viability of Quantum Non-Demolition (QND) logic gates. In simulation, information encoded in vortex helicity can be read analytically via continuous wake barometry without mechanically damping the particle's internal structure.

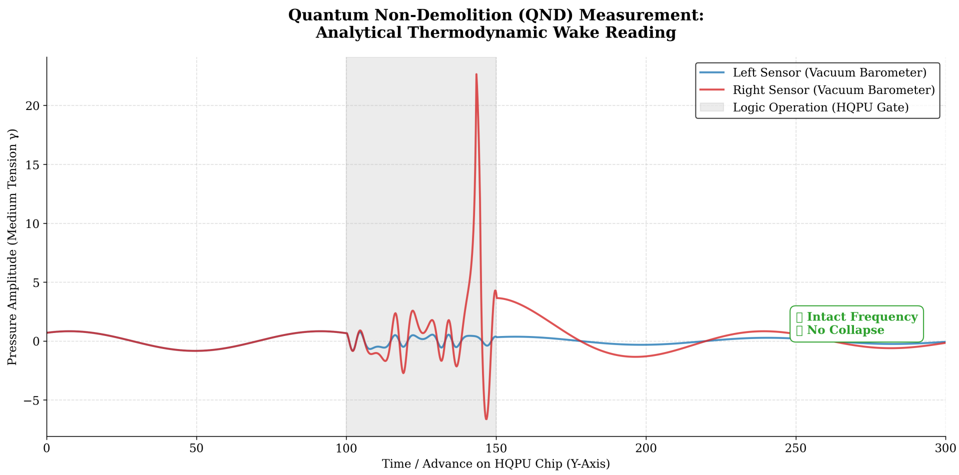


Figure 7: Simulated Non-Demolition measurement via wake barometry. Lateral pressure sensors read the encoded information without causing structural decoherence.

## 8 Conclusion

This paper presents the Deterministic Wave Engine as an exploratory hydrodynamic framework. By modeling a viscoelastic vacuum governed by an internal mesh tension ($\gamma_0$), we observe that wave-like statistical distributions, diffraction, and non-Markovian correlations can be simulated as emergent fluid-dynamic phenomena. While this work does not aim to definitively replace established quantum mechanics, it offers a robust computational analog that bridges deterministic local-realist N-body simulations with quantum-like behavior. The open-source availability of the DWE encourages further peer replication and investigation.

## References


[1] Couder, Y., & Fort, E. (2006). Single-Particle Diffraction and Interference at a Macroscopic Scale. *Physical Review Letters*, 97(15), 154101.

[2] Bush, J. W. M. (2015). Pilot-Wave Hydrodynamics. *Annual Review of Fluid Mechanics*, 47, 269–292.

[3] Volovik, G. E. (2003). *The Universe in a Helium Droplet*. Oxford University Press.

[4] Einstein, A., Podolsky, B., & Rosen, N. (1935). Can Quantum-Mechanical Description of Physical Reality Be Considered Complete? *Physical Review*, 47(10), 777.

[5] Bohm, D. (1952). A Suggested Interpretation of the Quantum Theory in Terms of "Hidden" Variables. I. *Physical Review*, 85(2), 166–179.

[6] Rozenman, Georgi Gary, et al. "Amplitude and phase of wave packets in a linear potential." *Physical Review Letters* 122.12 (2019): 124302.

[7] Rozenman, G. G., Shemer, L., & Arie, A. (2019). Quantum mechanical and optical analogies in surface gravity water waves. *Fluids*, 4(2), 96.

[8] Couto, F. B. (2026). Deterministic Wave Engine: A Hydrodynamic Computational Model of Wave-Particle Duality [Source Code]. GitHub.
`https://github.com/fbcouto/deterministic-wave-engine`

[9] Couto, F. B. (2026). Dissipative Gravitation Model: A Non-Linear Spatial Tension Approach to the N-Body Problem [Source Code]. GitHub.
`https://github.com/fbcouto/dissipative-gravitation-model`